Factorials can be generalised as double factorial as follows:

Double Factorial of n:

Factorial of an integer n, denoted by $n!$ can be viewed as a function $f : \mathbb{N} \cup \{0\} \to \mathbb{N}$, where $\mathbb{N}$ is the set of all Natural numbers, defined as

$$f(n) = \begin{cases} 1 & \text{for } n = 0, \\ n \times (n-1) \times (n-2) \times .... \times 3 \times 2 \times 1 & \text{for } n \neq 0. \end{cases}$$

One can define $n!!$ (double factorial of n) as

$$g(n) = \begin{cases} 1 & \text{for } n = 0, \\ n \times (n-2) \times (n-4) \times .... \times 4 \times 2 & \text{for n is even} \\ n \times (n-2) \times (n-4) \times .... \times 3 \times 1 & \text{for n is odd} \end{cases}$$

Accordingly, $5!! = 5 \times 3 \times 1 = 15$ and $8!! = 8 \times 6 \times 4 \times 2 = 384$.
Note that $n!! \neq (n!)!$ as $4!! = 8$ where as $(4!)! = (24)!$

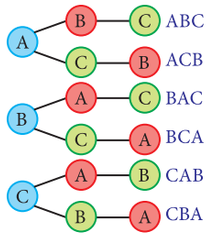
4.4 Permutations



What is a permutation ?

Permuations come in various disguises.

Suppose three friends A, B and C have to stand in line for a photograph. In how many order can they stand? Some of the possible arrangements (from left to right) are



A, B, C: A, C, B: B, A, C
B, C, A: C, B, A: C, A, B.

Thus there are six possible ways in which they can arrange themselves for the photograph.

Thus if 3 objects have to be arranged in a row there are $3 \times 2 \times 1 = 3!$ possible permutations. The number of permutations of 4 objects taken all at a time is $4 \times 3 \times 2 \times 1 = 4!$ Thus if n objects have to be arranged in a line there are $n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1 = n!$ possible arrangements or permutations.

Suppose you have 7 letters A,B,C,D,E,F and G. We want to make a 4 letter string. We have 7 choices for the 1st letter. Having chosen the first letter, we have 6 choices for the second letter. Proceeding this way, we have 4 choices for the 4th letter.
Hence, the number of permutations of 4 letters chosen from 7 letters is

$$7 \times 6 \times 5 \times 4 = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = \frac{7!}{3!} = \frac{7!}{(7-4)!}.$$

More generally, the number of distinct permutations of r objects which can be made from n distinct objects is $\frac{n!}{(n-r)!}$. It is denoted by ${}^nP_r$. The formal proof of this result will be proved in this section.